

CONTACT

monishnareshkumar@gmail.com

+65 8179 3493

Singapore

github.com/Brokenx25/

MONISH-PROJECTS

linkedin.com/in/

monish-naresh-kumar

brokenx25.github.io/portfolio_monish

DETAILS

NATIONALITY

Indian

WORK ELIGIBILITY

Student Pass — studying in Singapore

SKILLS

Python — Advanced

C / C++ / C# — Intermediate

SQL — Intermediate

Lua — Intermediate

HTML / CSS — Intermediate

Git — Advanced

ROS 2 · Webots · CoppeliaSim

OpenCV · MediaPipe · LLMs · 3D printing

LANGUAGES

English — Advanced (C1)

Tamil — Upper Int. (B2)

Hindi — Intermediate (B1)

Monish Naresh Kumar

ROBOTICS & AI ENGINEERING STUDENT

ABOUT ME

Robotics & AI engineering student (B.Eng., graduating September 2026) with a broad hands-on portfolio spanning perception, autonomous navigation, and human–robot interaction. Builds robots and intelligent systems end to end — from hardware assembly and control logic through computer-vision and speech pipelines — primarily in Python and C++ on ROS 2. Quick to learn new tools and effective in teams.

PROJECTS

PERCEPTION & MACHINE LEARNING

Computer Vision Pipeline Python · OpenCV · MediaPipe

- Built an end-to-end vision pipeline detecting faces, hand gestures, and body pose, integrating detection, tracking, and classification into one real-time system.

Speech Recognition Pipeline Python · speech & audio processing

- Developed a voice pipeline performing both speaker identification and speech-to-text, distinguishing who is speaking from what is being said.

AUTONOMOUS SYSTEMS & NAVIGATION

Autonomous Path-Following Car Python · path planning · sensor feedback

- Programmed a self-driving car to follow a planned route while sensing its surroundings and reacting in real time to obstacles, changes, and hazards.

Line-Following Robot — Built from Scratch C/C++ · microcontroller · IR sensors · 3D printing

- Designed and 3D-printed the chassis, assembled the wiring and control logic, and tuned the control loop for reliable autonomous line tracking.

Hazard Navigation Simulation CoppeliaSim · Lua · Python

- Modelled a real-world minefield scenario to analyse environmental dangers and plan safe navigation routes.

Multi-Behaviour Robot Simulation Webots · Python

- Implemented a simulated robot exhibiting 10 distinct autonomous behaviours across sensing, decision-making, and actuation.

HUMAN-ROBOT INTERACTION & SOCIAL ROBOTICS

Medical Robot Dog Assistant ROS 2 · Python · LLM · computer vision · Final-Year Project

- Developing an autonomous quadruped on ROS 2 that continuously monitors patients, integrating an LLM for natural interaction and a vision pipeline for perception — aimed as a lower-cost alternative to medical service dogs.

Teaching Assistant Robot Alpha Mini SDK · Python · Team Project

- Programmed an Alpha Mini humanoid, in a team of 4, to act as an interactive teaching assistant guiding students through learning difficulties.

Synchronized Robot Performance Alpha Mini · Python · Web (HTML/CSS) · Independent

- Coordinated multiple Alpha Mini robots to perform in perfect sync, triggered on demand via a QR-code-accessible web app.

SOFTWARE & SYSTEMS

Othello (Reversi) Game Python · Wireless Device Communication C/C++ · wireless protocols

- Built a fully playable Othello game (rules, move validation, turn logic) and implemented reliable wireless data exchange between two devices.

EDUCATION

Bachelor of Engineering in Robotics and AI

Expected Sep 2026

PSB Academy, Singapore

VOLUNTEERING

Event Volunteer, PSB Academy

2026

- Coordinated logistics and gathered participant feedback across campus events and programs to improve future delivery.